

Government Engineering College, Rajkot

Computer Engineering Department

Database Management System (BE03000091)

Academic Year 2026-27 (ODD 2026)

Assignment-1

No.	Question	CO
1.	Describe tasks performed by the Database Administrator.	CO1
2.	Describe the various views of data in a database system, including the physical, conceptual, and external levels.	CO1
3.	Explain DBMS System Architecture.	CO1
4	List the relational algebra operators. Discuss any four such algebra operators with suitable examples.	CO1
5	Consider the relational database given below. Give an expression in the relational algebra to express each of the following queries: Employee (person-name, street, city), Works (person-name, company-name, salary), Company (company-name, city), Manages (person-name, manager-name) (1) Find names of all employees. (2) Find the city of the employees whose name is 'jashu'. (3) Find the name and city of all employees who are having salary>50000. (4) Find the total salary of all employees who are working for company 'HCL'.	CO1

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Assignment-2

No.	Question	CO
1.	What is an integrity constraint? Explain primary key, reference key and check constraint with SQL syntax.	CO3
2	Consider the tables given below. Write the SQL queries for the questions given below: T1 (Empno, Ename , Salary, Designation,) T2 (Empno, Deptno.) (1) Display all the details of the employee whose salary is lesser than 10000. (2) Display the Deptno in which Employees with name starting with letter 'S' is working. (3) Add a new column Deptname in table T2. (4) Change the designation of Geeta from 'Manager' to 'Senior Manager'. (5) Find the total salary of all the employees department wise. (6) Add Empno as primary key in existing table T1. (7) Display the Deptno having the highest number of employees.	CO3
3	Explain types of Join in MySQL with examples.	CO3
4	Draw an E-R diagram for the College management system.	CO2
5	What is an Entity-Relationship (E-R) model? Describe the key components of an E-R diagram: entities, attributes, and relationships.	CO2